

LOGISTICS AND PLANNING STRATEGIES

Activities of Logistics

Logistics involves all activities required to move and store goods efficiently from source to consumption.

Core Activities

- 1.Customer Service** – Ensuring correct product delivery, accuracy, speed, responsiveness.
- 2.Inventory Management** – Forecasting demand, ordering, holding stock.
- 3.Transportation** – Selecting mode (road/rail/air/sea), routing, freight consolidation.
- 4.Warehousing** – Receiving, storing, picking, packing, dispatching goods.
- 5.Order Processing** – Order entry → verification → invoicing → dispatch.
- 6.Material Handling** – Movement within warehouse via forklifts, conveyors, robotics.
- 7.Packaging** – Protecting goods, unitization, labeling.
- 8.Facilities Management** – Managing distribution centers and transportation hubs.
- 9.Reverse Logistics** – Handling returns, recycling, disposal.

Aims of Logistics

- Right product
- Right quantity
- Right condition
- Right place
- Right time
- Right customer
- Right cost

Additional Aims

- Minimize total logistics cost
- Reduce delivery lead time
- Maximize asset utilization
- Improve customer satisfaction
- Maintain quality throughout the supply chain

Importance

Competitive Advantage

Cost Efficiency

Customer Satisfaction

Global Trade Enablement

Economic Contribution

Supply Chain Integration

Explanation

Fast delivery, low cost, reliability attract customers

Efficient logistics reduces transportation, inventory costs

Accurate and on-time deliveries

Connects suppliers and markets around the world

Creates jobs, supports industries

Smooth flow across procurement → production → distribution

Current Trends in Logistics

- 1.Digitalization** – AI, IoT, cloud logistics, sensors on trucks.
- 2.Automation** – Robotics, automated picking, autonomous forklifts.
- 3.Green Logistics** – Electric trucks, route optimization, recyclable packaging.
- 4.Omni-Channel Distribution** – Integrated online + offline supply chains.
- 5.Last Mile Innovations** – Drones, lockers, micro-fulfillment centers.
- 6.Blockchain** – Secure tracking, smart contracts.
- 7.Real-Time Visibility** – GPS tracking, digital twins.
- 8.Collaboration & 4PL** – Outsourcing logistics to expert providers.

LOGISTICS STRATEGY

Strategic Decisions in Logistics

Strategic decisions are **long-term**, high-level decisions affecting the entire supply chain.

Major Strategic Areas

1. **Facility Location** – Where to place warehouses/plants.
2. **Capacity Management** – Long-term resource planning.
3. **Transportation Network** – Mode choices, fleet size, routing.
4. **Inventory Policy** – Stock levels, safety stock, reorder points.
5. **Outsourcing vs In-house** – Using 3PL/4PL providers.
6. **Technology Selection** – WMS, TMS, automation.
7. **Customer Service Standards** – Desired service levels.

Logistics Strategy

Logistics strategy aligns logistics activities with overall business strategy.

Components

- **Cost Leadership Strategy** → minimize logistics cost.
- **Differentiation Strategy** → focus on speed, customization.
- **Responsiveness Strategy** → agile & flexible supply chain.
- **Lean Strategy** → reduce waste, optimize flows.
- **Green Strategy** → sustainability driven.

Designing a Logistics Strategy

Steps:

- 1. Analyze Business Goals** – market, customers, competition.
- 2. Assess Current Logistics Performance** – gaps, process mapping.
- 3. Forecast Future Logistics Needs** – volumes, lead times, customer expectations.
- 4. Select Key Logistics Drivers** – location, inventory, transport, technology.
- 5. Design Network Structure** – number, size, placement of facilities.
- 6. Develop Implementation Plan** – budgets, training, KPIs.
- 7. Monitor & Improve** – continuous improvement loop.

LOCATING FACILITIES

Importance of Location

Facility location affects:

- Customer service levels
- Transportation cost
- Operating cost
- Delivery time
- Access to labor, suppliers, markets
- Risk due to natural disasters or political issues

Choosing the Geographic Region

Factors:

- Proximity to customers
- Availability of labor
- Land cost
- Infrastructure (roads, ports, airports)
- Regulations & taxes
- Utility availability
- Political stability

Infinite Set Approaches

Used when facility can be placed *anywhere* on a plane or region.

Examples:

Center of Gravity Method (COG)

Calculates best location to minimize weighted distance.

Formula:

$$X = \frac{\sum (x_i \cdot w_i)}{\sum w_i}$$

$$Y = \frac{\sum (y_i \cdot w_i)}{\sum w_i}$$

Weber Problem

Minimizing weighted distances for multiple supply & demand points.

Feasible Set Approaches

Used when location must be chosen from a set of limited choices (cities, regions).

Methods:

- Factor Rating Method
- Cost-Volume Analysis
- Break-even Analysis
- Scoring Models
- Weighted Point System

Network Models

Used for evaluating distribution networks.

1. Transportation Model

Finds the least-cost shipping plan from origins to destinations.

2. Transshipment Model

Allows movement through intermediate points.

3. Assignment Model

Assigns tasks (or warehouses) to locations optimally.

Location Planning

Steps:

1. Identify criteria
2. Develop location alternatives
3. Gather data
4. Apply quantitative and qualitative models
5. Evaluate costs, service levels, risks
6. Select best site
7. Validate via simulation or pilot operation

PLANNING RESOURCES

Types of Planning

Planning Type	Time Horizon	Purpose
Strategic Planning	2–10 years	Long-term network and capacity decisions
Tactical Planning	6–24 months	Resource allocation, inventory policies
Operational Planning	Daily–Weekly	Scheduling, dispatching, routing

Capacity Planning

Ensures the company has enough **resources** (machines, labor, warehouses, vehicles) to meet demand.

Steps:

- 1.Forecast demand
- 2.Estimate required capacity
- 3.Identify capacity gaps
- 4.Develop capacity alternatives
- 5.Evaluate costs and choose best option

Adjusting Capacity

Ways to Increase Capacity

- Add equipment/automation
- Hire more labor
- Add shifts (e.g., night shift)
- Overtime
- Outsourcing or subcontracting
- Expand facility
- Cross-training employees

Ways to Decrease Capacity

- Reduce shifts
- Temporary layoffs
- Idle machines
- Outsource temporarily
- Combine tasks

Tactical Planning

Tactical planning aligns medium-term resources with expected demand.

Focus areas:

- Production planning
- Inventory levels
- Transportation contracts
- Workforce planning
- Seasonal adjustments
- Purchasing and supplier scheduling

Schedules

Schedules are detailed plans for allocating resources over short periods.

Types:

- Master Production Schedule (MPS)
- Employee Work Schedules
- Vehicle Routing Schedules
- Warehouse Operation Schedules (pick/pack)
- Maintenance Schedules

A good schedule improves:

- Resource utilization
- On-time delivery
- Cost control
- Flexibility